SC573

DSP

File: sc573.kicad_sch

power

Power Controller and Regulators

File: power.kicad_sch

Ethernet

Ethernet Phy

File: ethernet.kicad_sch

PoE

PoE

File: poe.kicad_sch

usb

USB

File: usb.kicad_sch

IO_Connectors

IO Connectors

File: io_connectors.kicad_sch

extern_memory

Extern Memory

File: extern_memory.kicad_sch

Sheet: /
File: cpu_board.kicad_sch

Title:

Size: A4 | Date: | **Rev:**

KiCad E.D.A. 10.0.6 | Id: 1/15

dsp_ports

File: dsp_ports.kicad_sch

sc573_power

File: sc573_power.kicad_sch

sc573_system

File: sc573_system.kicad_sch

ddr

File: ddr.kicad_sch

bypass_caps

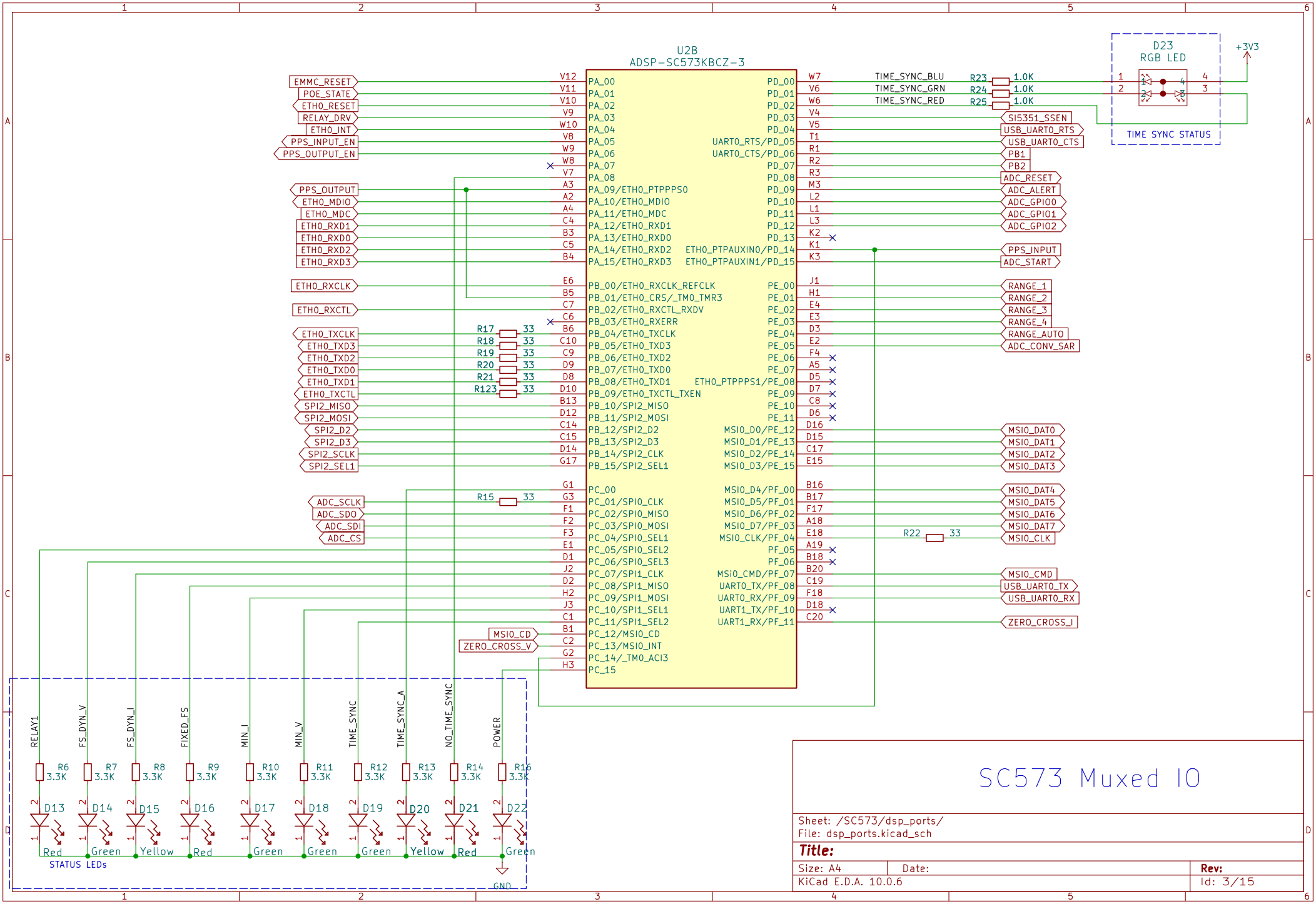
File: bypass_caps.kicad_sch

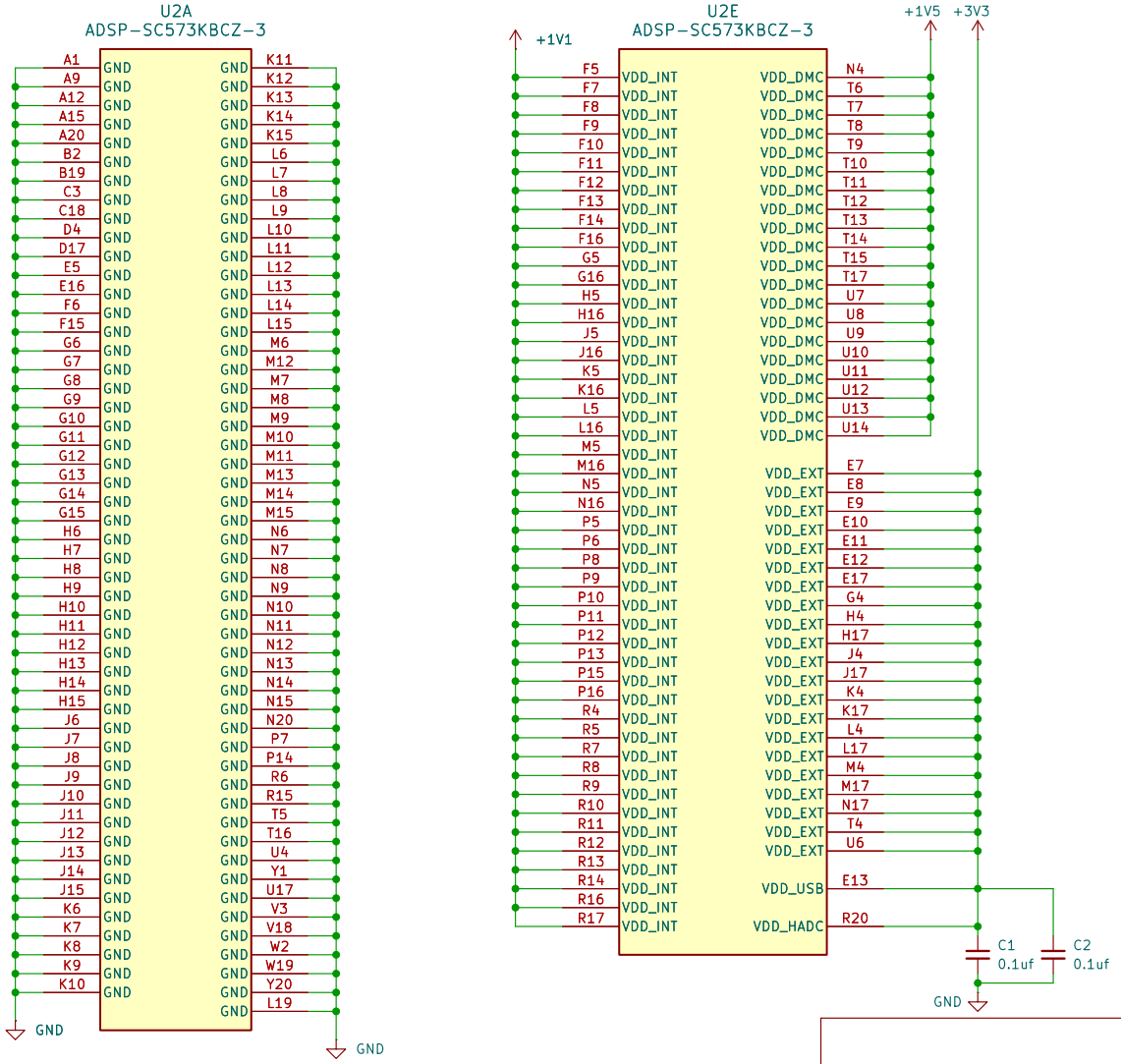
DSP

Sheet: /SC573/
File: sc573.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 2/15


U2B
ADSP-SC573KBCZ-3
D23
RGB LED
TIME SYNC STATUS
+3V3
STATUS LEDs
GND
SC573 Muxed IO
Sheet: /SC573/dsp_ports/
File: dsp_ports.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 10.0.6
Id: 3/15

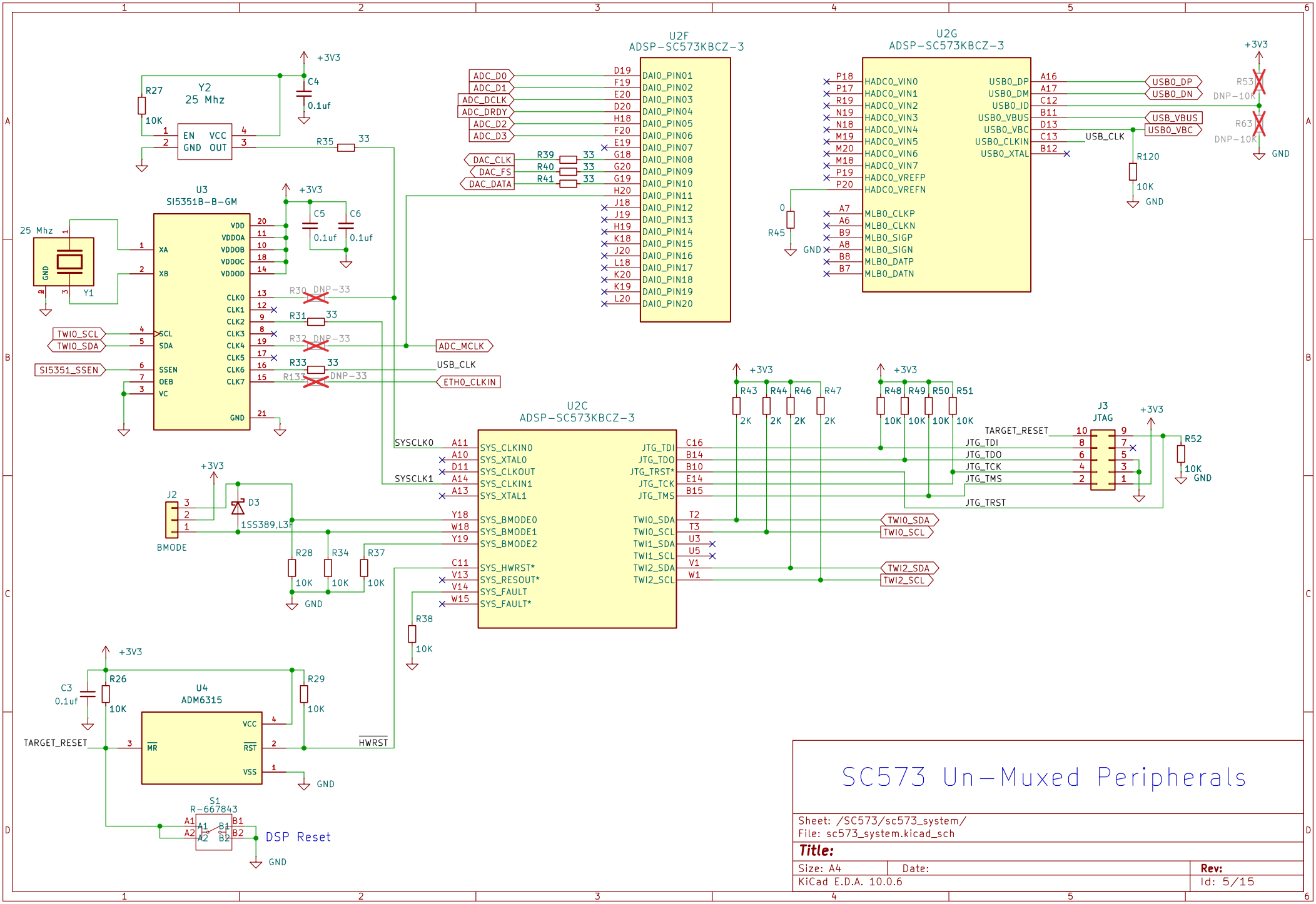


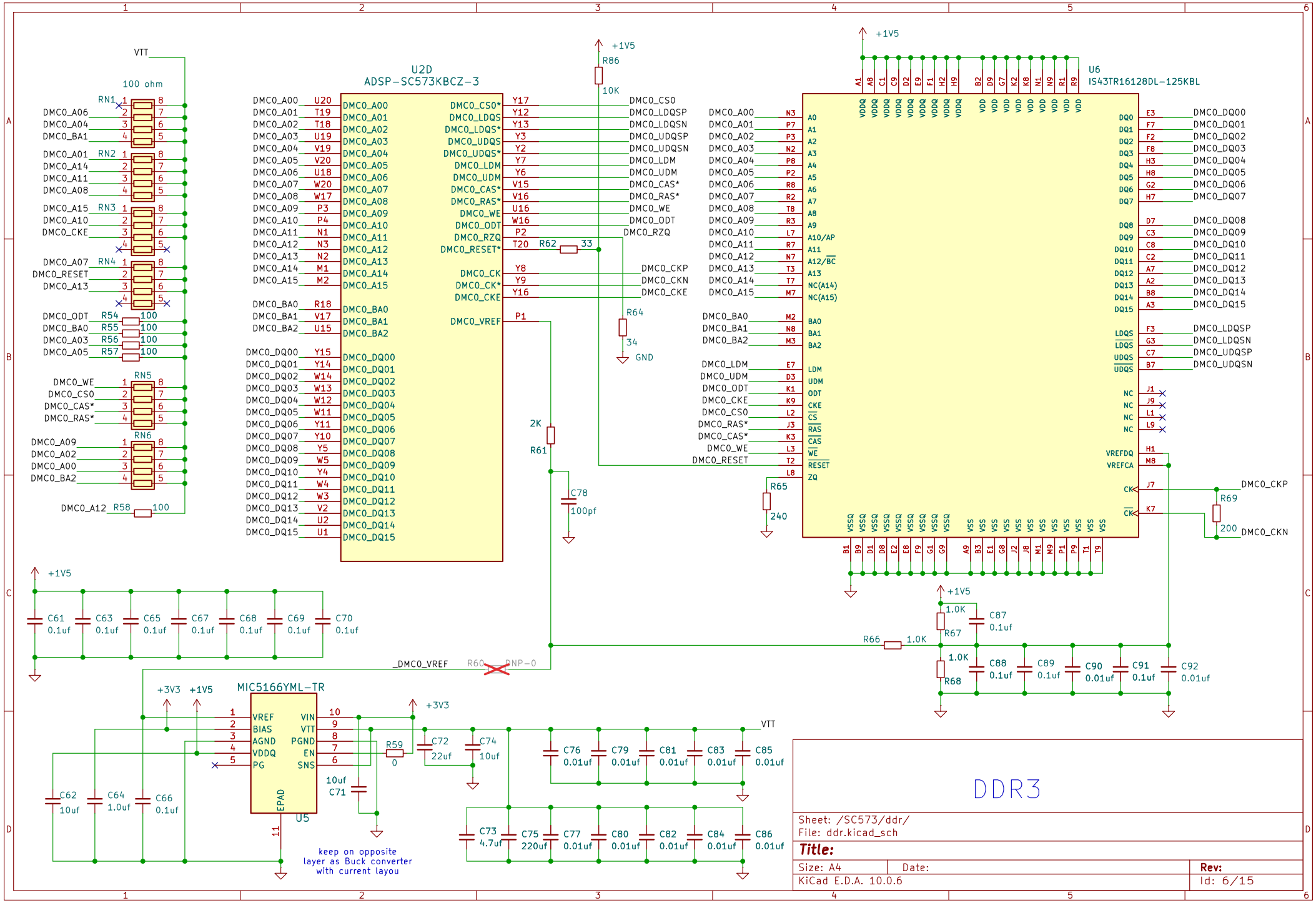
DSP Power

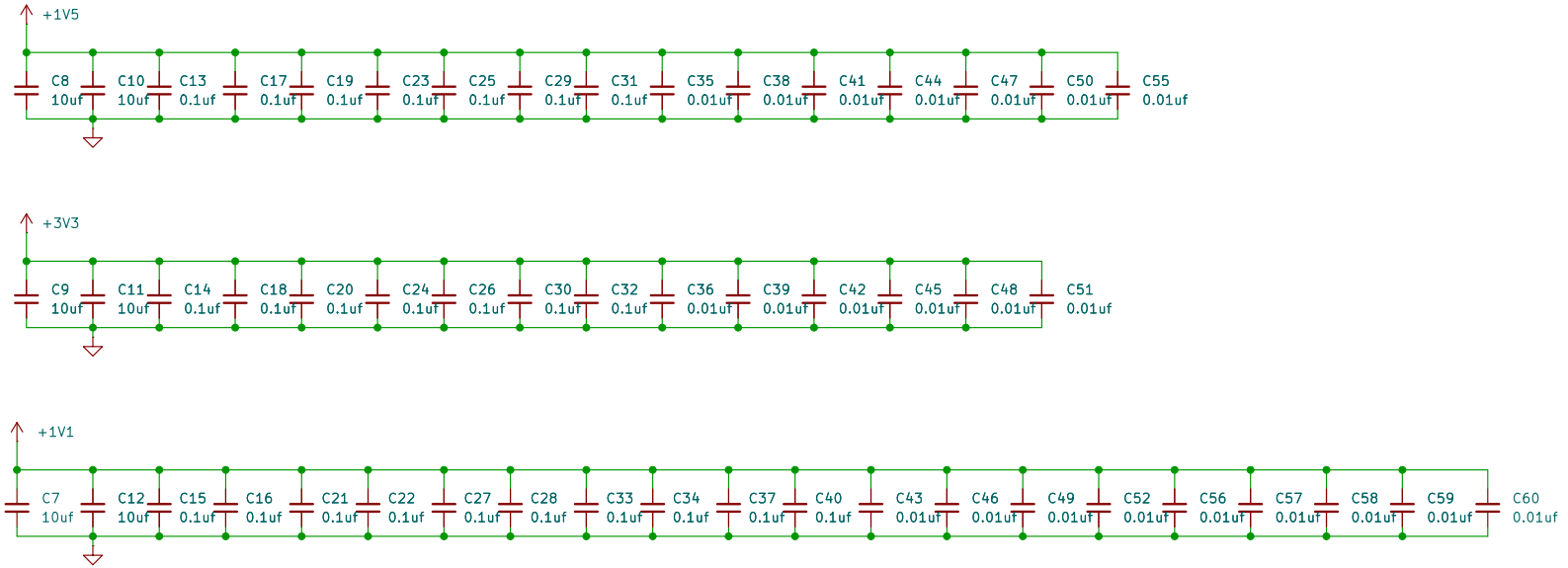
Sheet: /SC573/sc573_power/
File: sc573_power.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 4/15


U2F
ADSP-SC573KBCZ-3
U2G
ADSP-SC573KBCZ-3
U3
SI5351B-B-GM
U2C
ADSP-SC573KBCZ-3
J3
JTAG
U4
ADM6315
S1
R-667843
DSP Reset
Y2
25 Mhz
BMODE
SC573 Un-Muxed Peripherals
Sheet: /SC573/sc573_system/
File: sc573_system.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 10.0.6
Id: 5/15


U2D
ADSP-SC573KBCZ-3
U6
IS43TR16128DL-125KBL
MIC5166YML-TR
U5
DDR3
Sheet: /SC573/ddr/
File: ddr.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 10.0.6
Id: 6/15
keep on opposite layer as Buck converter with current layou
100 ohm
VTT
+1V5
+3V3
GND
DMC0_CKP
DMC0_CKN
_DMC0_VREF
DNP-0

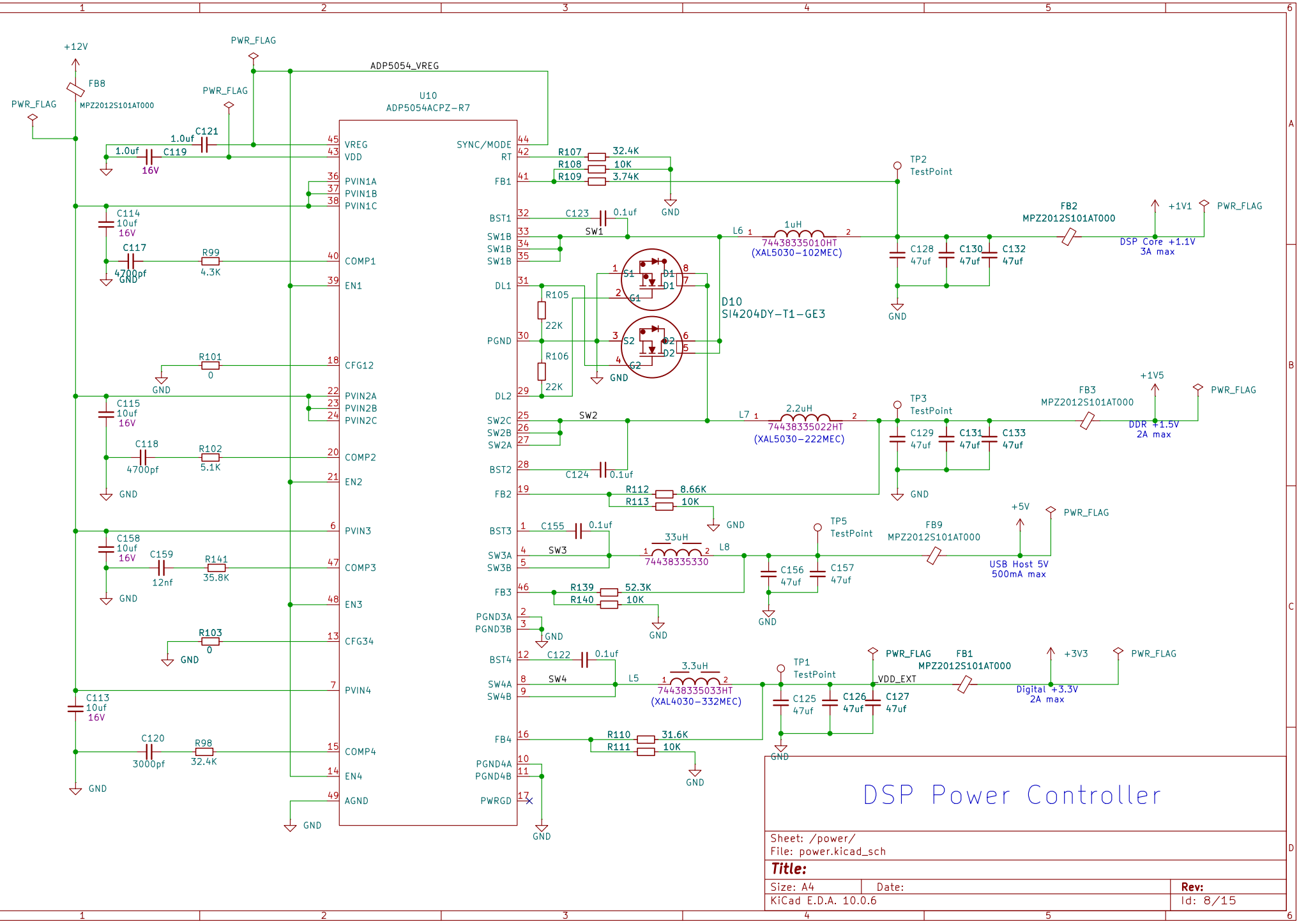


DSP Bypass Caps

Sheet: /SC573/bypass_caps/
File: bypass_caps.kicad_sch

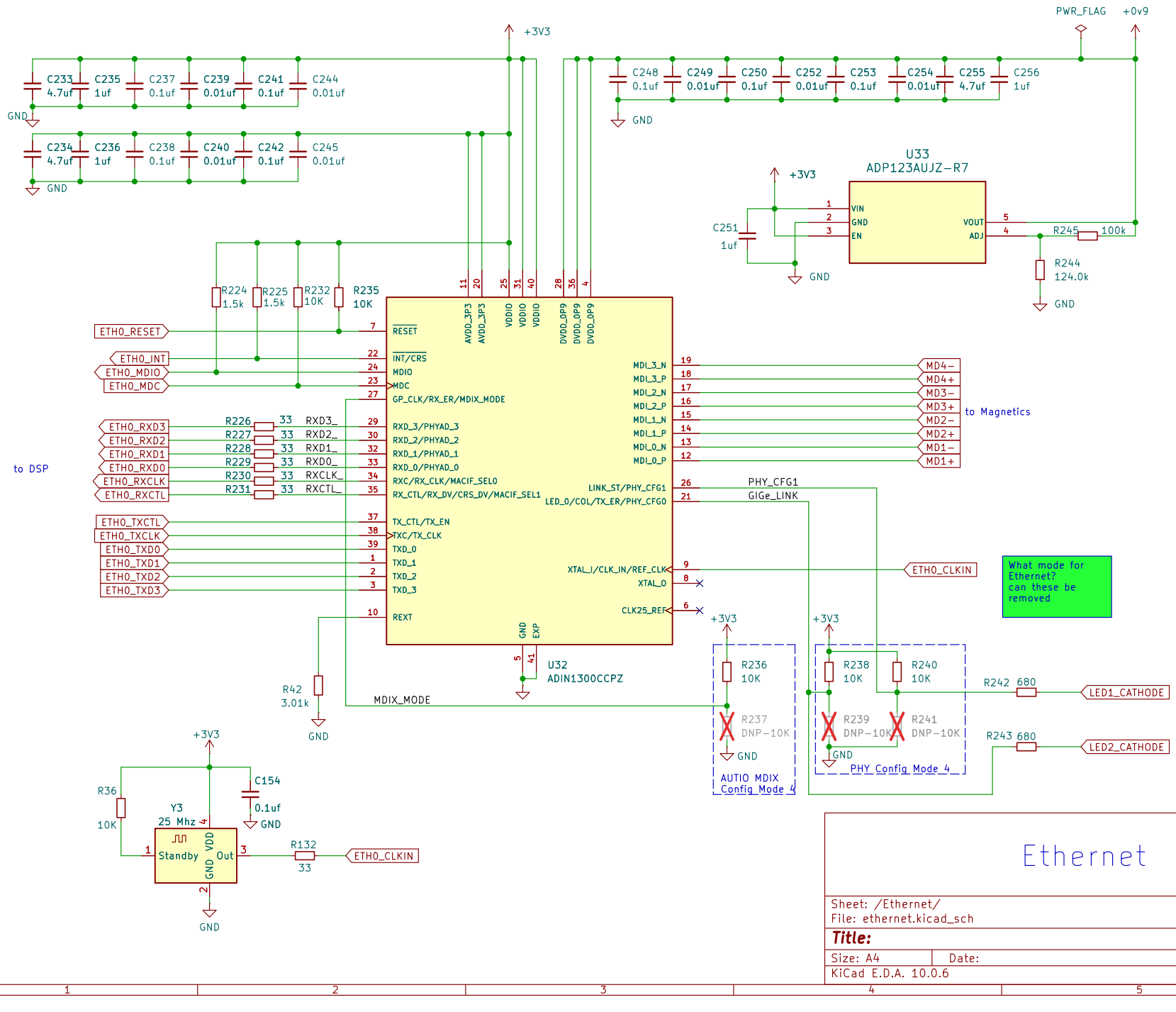
Title:

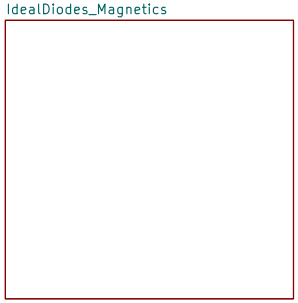
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 7/15

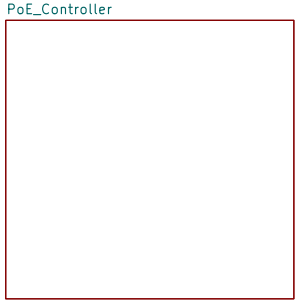


DSP Power Controller

Sheet: /power/		
File: power.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 8/15


PWR_FLAG
+0v9
+3V3
C233 4.7uf
C235 1uf
C237 0.1uf
C239 0.01uf
C241 0.1uf
C244 0.01uf
C234 4.7uf
C236 1uf
C238 0.1uf
C240 0.01uf
C242 0.1uf
C245 0.01uf
GND
C248 0.1uf
C249 0.01uf
C250 0.1uf
C252 0.01uf
C253 0.1uf
C254 0.01uf
C255 4.7uf
C256 1uf
U33
ADP123AUJZ-R7
VIN
GND
EN
VOUT
ADJ
C251 1uf
R245 100k
R244 124.0k
R224 1.5k
R225 1.5k
R232 10K
R235 10K
AVDD_3P3
VDDIO
DVDD_0P9
RESET
INT/CRS
MDIO
MDC
GP_CLK/RX_ER/MDIX_MODE
RXD_3/PHYAD_3
RXD_2/PHYAD_2
RXD_1/PHYAD_1
RXD_0/PHYAD_0
RXC/RX_CLK/MACIF_SEL0
RX_CTL/RX_DV/CRS_DV/MACIF_SEL1
TX_CTL/TX_EN
TXC/TX_CLK
TXD_0
TXD_1
TXD_2
TXD_3
REXT
MDI_3_N
MDI_3_P
MDI_2_N
MDI_2_P
MDI_1_N
MDI_1_P
MDI_0_N
MDI_0_P
LINK_ST/PHY_CFG1
LED_0/COL/TX_ER/PHY_CFG0
XTAL_I/CLK_IN/REF_CLK
XTAL_O
CLK25_REF
EXP
U32
ADIN1300CCPZ
ETH0_RESET
ETH0_INT
ETH0_MDIO
ETH0_MDC
ETH0_RXD3
ETH0_RXD2
ETH0_RXD1
ETH0_RXD0
ETH0_RXCLK
ETH0_RXCTL
ETH0_TXCTL
ETH0_TXCLK
ETH0_TXD0
ETH0_TXD1
ETH0_TXD2
ETH0_TXD3
R226 33 RXD3_
R227 33 RXD2_
R228 33 RXD1_
R229 33 RXD0_
R230 33 RXCLK_
R231 33 RXCTL_
to DSP
MD4−
MD4+
MD3−
MD3+
MD2−
MD2+
MD1−
MD1+
to Magnetics
PHY_CFG1
GIGe_LINK
ETH0_CLKIN
What mode for Ethernet? can these be removed
R42 3.01k
MDIX_MODE
R236 10K
R237 DNP-10K
AUTIO MDIX Config Mode 4
R238 10K
R240 10K
R239 DNP-10K
R241 DNP-10K
PHY Config Mode 4
R242 680
R243 680
LED1_CATHODE
LED2_CATHODE
R36 10K
Y3
25 Mhz
Standby
VDD
GND
Out
C154 0.1uf
R132 33
ETH0_CLKIN
Ethernet
Sheet: /Ethernet/
File: ethernet.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 10.0.6
Id: 11/15



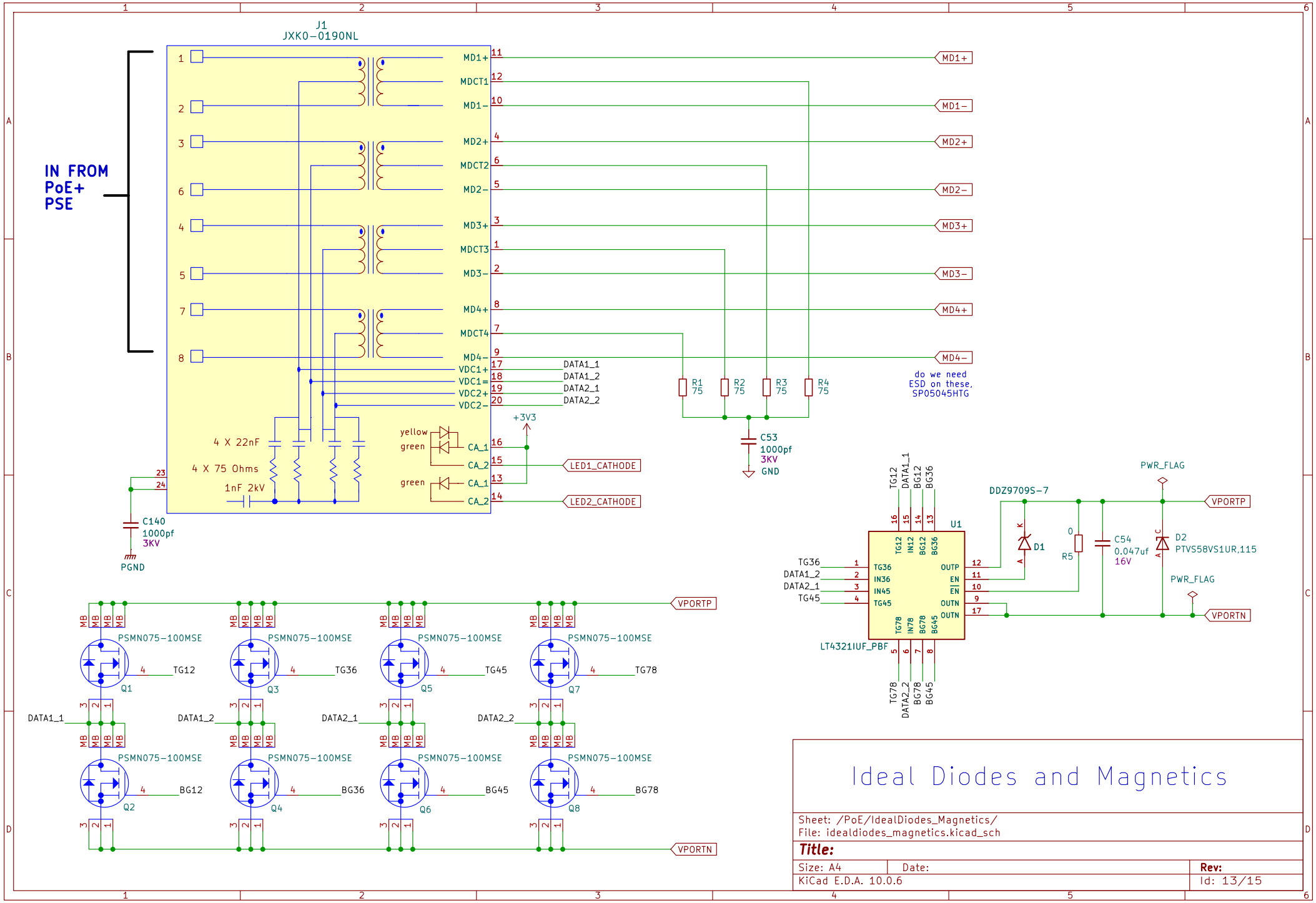


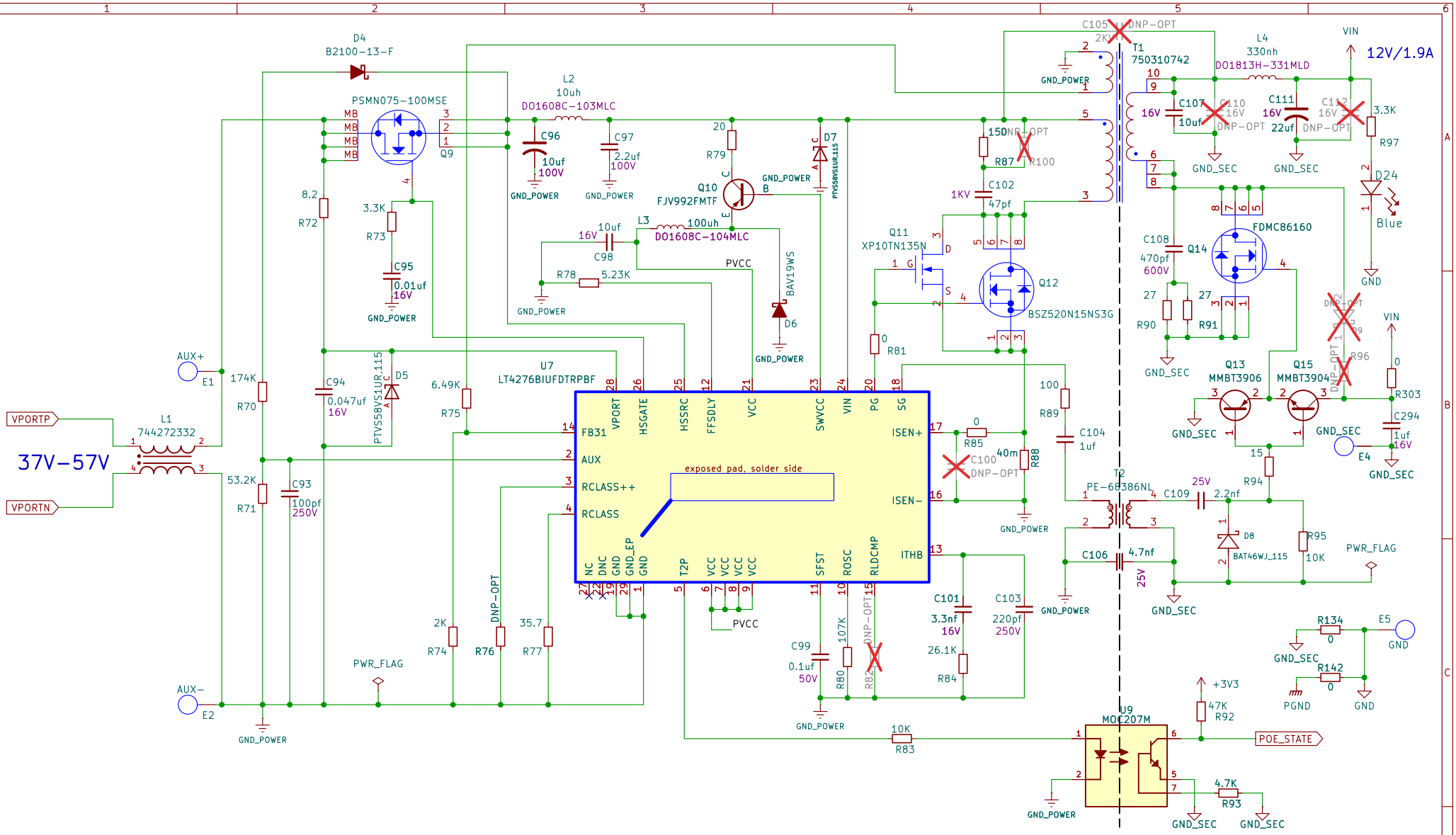
PoE

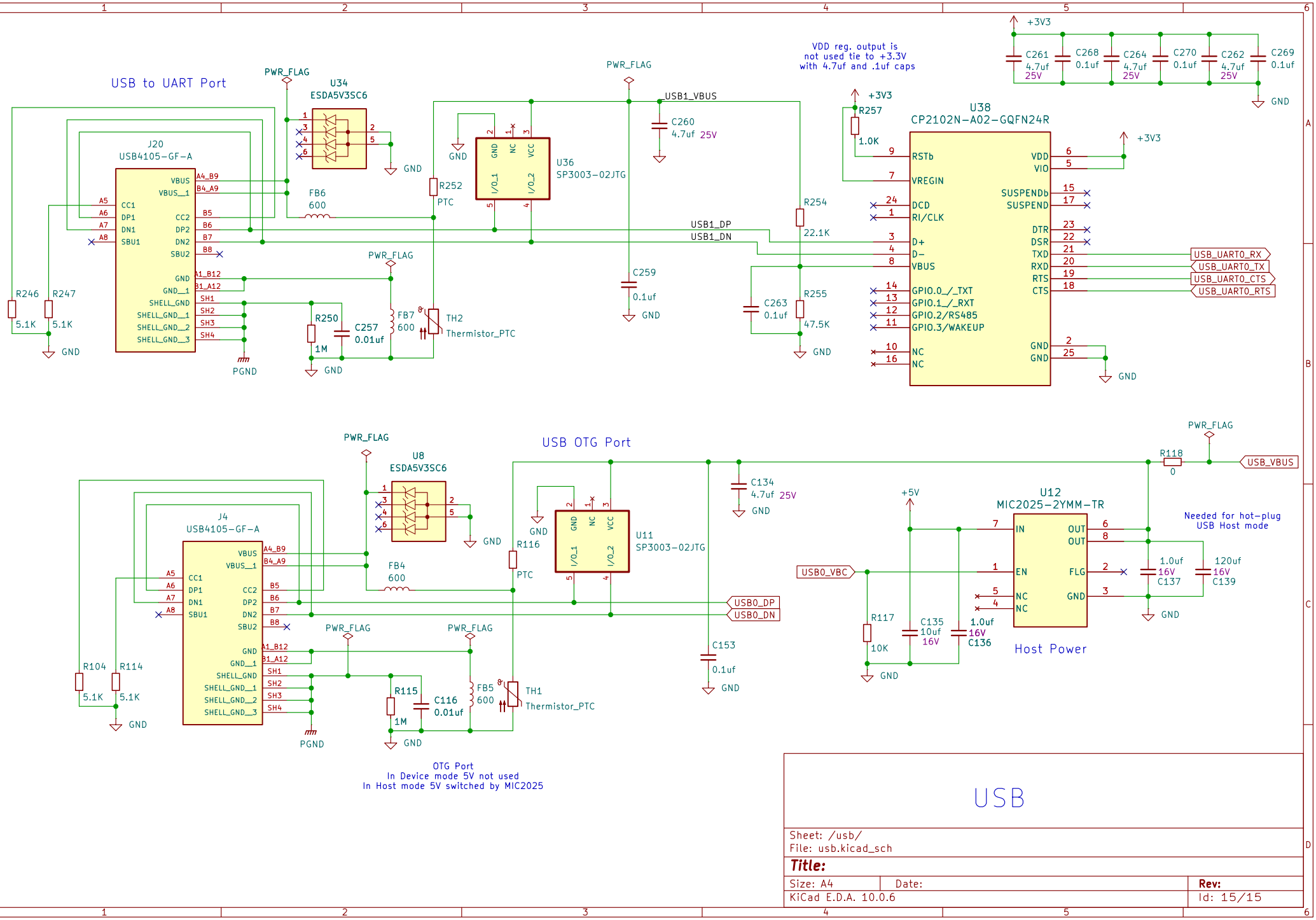
Sheet: /PoE/
File: poe.kicad_sch

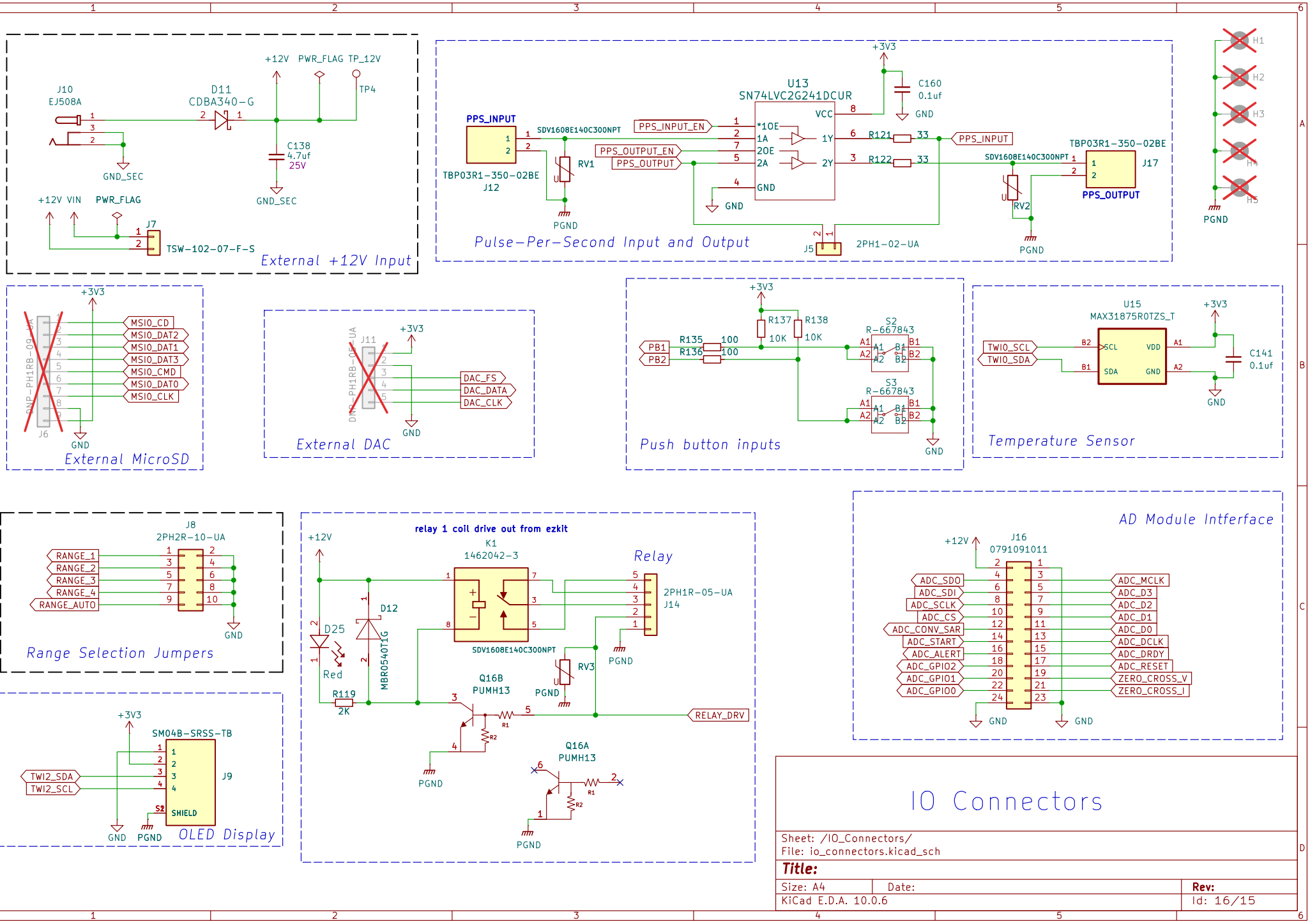
Title:

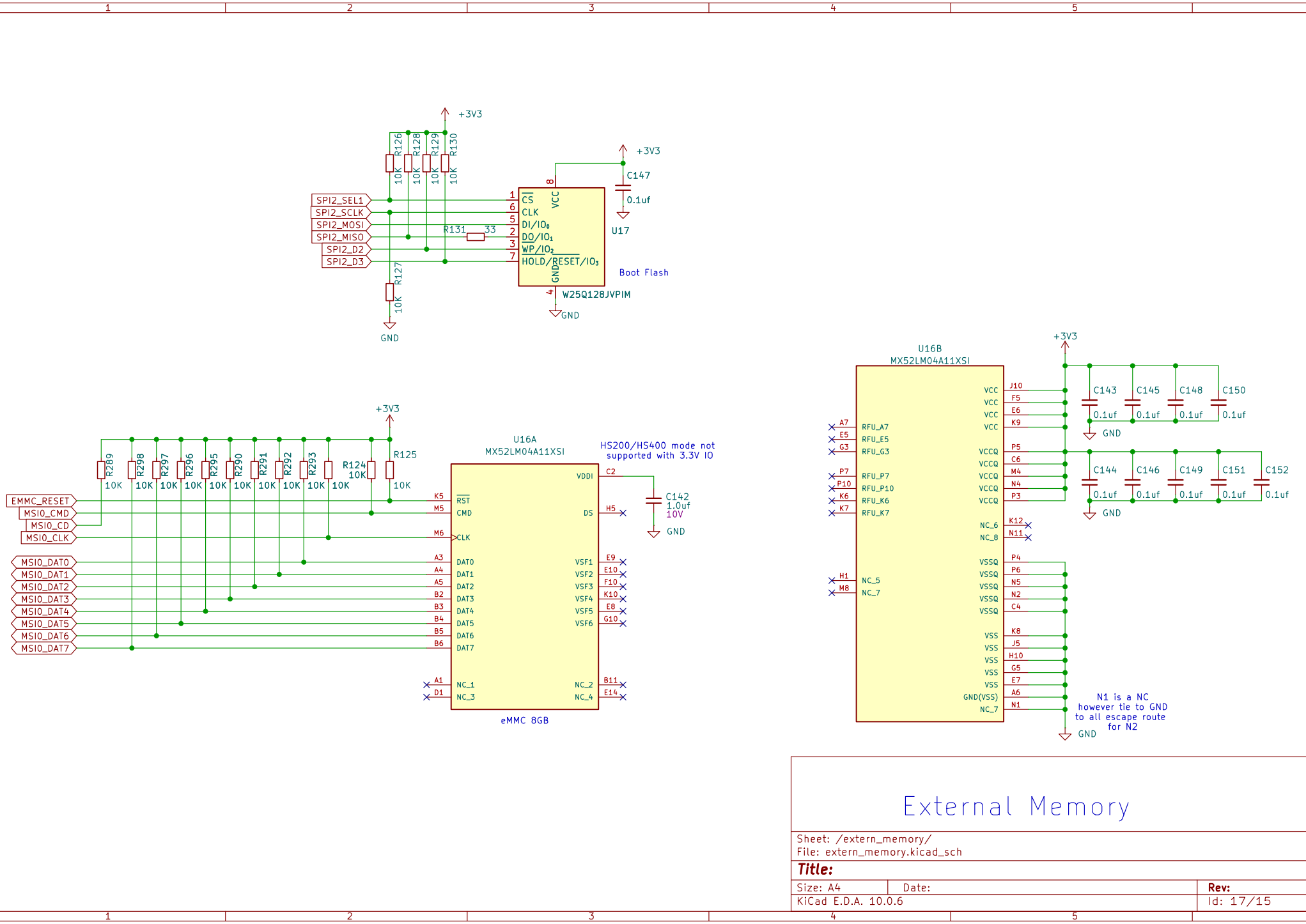
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 12/15


J1
JXK0-0190NL
IN FROM
PoE+
PSE
MD1+
MDCT1
MD1-
MD2+
MDCT2
MD2-
MD3+
MDCT3
MD3-
MD4+
MDCT4
MD4-
VDC1+
VDC1=
VDC2+
VDC2-
DATA1_1
DATA1_2
DATA2_1
DATA2_2
4 X 22nF
4 X 75 Ohms
1nF 2kV
yellow
green
green
CA_1
CA_2
CA_1
CA_2
+3V3
LED1_CATHODE
LED2_CATHODE
R1 75
R2 75
R3 75
R4 75
do we need
ESD on these,
SP05045HTG
C53
1000pf
3KV
GND
C140
1000pf
3KV
PGND
PSMN075-100MSE
Q1
Q2
Q3
Q4
Q5
Q6
Q7
Q8
TG12
BG12
TG36
BG36
TG45
BG45
TG78
BG78
VPORTP
VPORTN
U1
LT4321IUF_PBF
DDZ9709S-7
D1
R5
0
C54
0.047uf
16V
D2
PTVS58VS1UR,115
PWR_FLAG
PWR_FLAG
Ideal Diodes and Magnetics
Sheet: /PoE/IdealDiodes_Magnetics/
File: idealdiodes_magnetics.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 10.0.6
Id: 13/15


D4
B2100-13-F
PSMN075-100MSE
L2
10uh
DO1608C-103MLC
C96
10uf
100V
C97
2.2uf
100V
Q9
Q10
FJV992FMTF
R79
20
D7
PTVS58VS1UR,115
L3
100uh
DO1608C-104MLC
C98
10uf
16V
R78
5.23K
PVCC
BAV19WS
D6
R72
8.2
R73
3.3K
C95
0.01uf
16V
AUX+
E1
174K
R70
C94
0.047uf
16V
D5
PTVS58VS1UR,115
6.49K
R75
U7
LT4276BIUFDTRPBF
exposed pad, solder side
FB31
AUX
RCLASS++
RCLASS
NC
DNC
GND
GND_EP
GND
VPORT
HSGATE
HSSRC
FFSDLY
VCC
SWVCC
VIN
PG
SG
ISEN+
ISEN-
ITHB
T2P
VCC
SFST
ROSC
RLDCMP
VPORTP
VPORTN
37V-57V
L1
744272332
53.2K
R71
C93
100pf
250V
2K
R74
DNP-OPT
R76
35.7
R77
PWR_FLAG
AUX-
E2
PVCC
C99
0.1uf
50V
107K
R80
DNP-OPT
R82
C101
3.3nf
16V
26.1K
R84
C103
220pf
250V
10K
R83
Q11
XP10TN135N
Q12
BSZ520N15NS3G
R87
150
DNP-OPT
R100
C102
1KV
47pf
R81
0
R85
0
C100
DNP-OPT
40m
R88
100
R89
C104
1uf
C105
2KV
DNP-OPT
T1
750310742
L4
330nh
DO1813H-331MLD
C107
16V
10uf
C110
16V
DNP-OPT
C111
16V
22uf
C112
16V
DNP-OPT
3.3K
R97
D24
Blue
VIN
12V/1.9A
GND_SEC
GND_POWER
C108
470pf
600V
Q14
FDMC86160
R90
27
R91
27
Q13
MMBT3906
Q15
MMBT3904
R96
DNP-OPT
R303
0
C294
1uf
16V
E4
T2
PE-68386NL
C109
25V
2.2nf
R94
D8
BAT46WJ_115
R95
10K
C106
4.7nf
25V
R134
0
E5
GND
R142
0
PGND
U9
MOC207M
+3V3
47K
R92
POE_STATE
4.7K
R93
PoE Controller
Sheet: /PoE/PoE_Controller/
File: poe_controller.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 10.0.6
Id: 14/15


USB to UART Port
J20
USB4105-GF-A
U34
ESDA5V3SC6
U36
SP3003-02JTG
U38
CP2102N-A02-GQFN24R
VDD reg. output is
not used tie to +3.3V
with 4.7uf and .1uf caps
+3V3
C261 4.7uf 25V
C268 0.1uf
C264 4.7uf 25V
C270 0.1uf
C262 4.7uf 25V
C269 0.1uf
R257 1.0K
C260 4.7uf 25V
USB1_VBUS
USB1_DP
USB1_DN
R254 22.1K
R255 47.5K
C263 0.1uf
C259 0.1uf
R252 PTC
FB6 600
R246 5.1K
R247 5.1K
R250 1M
C257 0.01uf
FB7 600
TH2 Thermistor_PTC
PWR_FLAG
GND
PGND
USB_UART0_RX
USB_UART0_TX
USB_UART0_CTS
USB_UART0_RTS
USB OTG Port
J4
USB4105-GF-A
U8
ESDA5V3SC6
U11
SP3003-02JTG
U12
MIC2025-2YMM-TR
R118 0
USB_VBUS
C134 4.7uf 25V
C153 0.1uf
R116 PTC
FB4 600
R104 5.1K
R114 5.1K
R115 1M
C116 0.01uf
FB5 600
TH1 Thermistor_PTC
USB0_DP
USB0_DN
USB0_VBC
R117 10K
C135 10uf 16V
C136 1.0uf 16V
C137 1.0uf 16V
C139 120uf 16V
+5V
Needed for hot-plug
USB Host mode
Host Power
OTG Port
In Device mode 5V not used
In Host mode 5V switched by MIC2025
USB
Sheet: /usb/
File: usb.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 10.0.6
Id: 15/15


External +12V Input
J10
EJ508A
D11
CDBA340-G
+12V
PWR_FLAG
TP_12V
TP4
C138
4.7uf
25V
GND_SEC
+12V VIN
PWR_FLAG
J7
TSW-102-07-F-S
Pulse-Per-Second Input and Output
U13
SN74LVC2G241DCUR
+3V3
C160
0.1uf
GND
PPS_INPUT
SDV1608E140C300NPT
TBP03R1-350-02BE
J12
RV1
PGND
PPS_INPUT_EN
PPS_OUTPUT_EN
PPS_OUTPUT
*1OE
1A
2OE
2A
VCC
1Y
2Y
R121
R122
33
PPS_INPUT
TBP03R1-350-02BE
J17
PPS_OUTPUT
RV2
J5
2PH1-02-UA
H1
H2
H3
H4
H5
External MicroSD
DNP-PH1RB-09-UA
J6
MSI0_CD
MSI0_DAT2
MSI0_DAT1
MSI0_DAT3
MSI0_CMD
MSI0_DAT0
MSI0_CLK
External DAC
DNP-PH1RB-05-UA
J11
DAC_FS
DAC_DATA
DAC_CLK
Push button inputs
R137
R138
10K
PB1
PB2
R135
R136
100
S2
S3
R-667843
Temperature Sensor
U15
MAX31875R0TZS_T
TWI0_SCL
TWI0_SDA
SCL
SDA
VDD
GND
C141
0.1uf
Range Selection Jumpers
J8
2PH2R-10-UA
RANGE_1
RANGE_2
RANGE_3
RANGE_4
RANGE_AUTO
OLED Display
SM04B-SRSS-TB
J9
TWI2_SDA
TWI2_SCL
SHIELD
relay 1 coil drive out from ezkit
K1
1462042-3
Relay
2PH1R-05-UA
J14
SDV1608E140C300NPT
D12
MBR0540T1G
D25
Red
R119
2K
RV3
Q16B
PUMH13
Q16A
PUMH13
R1
R2
RELAY_DRV
AD Module Intferface
J16
0791091011
ADC_SDO
ADC_SDI
ADC_SCLK
ADC_CS
ADC_CONV_SAR
ADC_START
ADC_ALERT
ADC_GPIO2
ADC_GPIO1
ADC_GPIO0
ADC_MCLK
ADC_D3
ADC_D2
ADC_D1
ADC_D0
ADC_DCLK
ADC_DRDY
ADC_RESET
ZERO_CROSS_V
ZERO_CROSS_I
IO Connectors
Sheet: /IO_Connectors/
File: io_connectors.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 10.0.6
Id: 16/15



External Memory

Sheet: /extern_memory/		
File: extern_memory.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 17/15